

DANIEL GUO

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Education

University of California, Los Angeles (UCLA) | *Computer Science* Sep. 2023 – Expected June 2027

- **B.S. Computer Science** (Expected June 2026), **M.S. Computer Science** (Expected June 2027) **GPA: 3.81**
- **Relevant Coursework:** Data Science, Neural Networks and Deep Learning I and II, Optimization
- Data Structures and Algorithms, Software Construction, Operating Systems, Computer Graphics/Animation

Experience

Walmart Global Tech | *Software Intern* June 2025 – August 2025

- Developed local end-to-end pipeline for AI-generated code detection with **PyTorch** and **scikit-learn**, processing **200K+ LOC** per run under strict memory constraints; enabled fine-tuning and inference on BERT, LLaMA models
- Automated **synthetic data generation** via LLMs and implemented **sliding window inference** for long-sequence classification; built interactive evaluation dashboard using **JavaScript** and **D3.js** with **CosmosDB** backend

Trustworthy AI Lab | *Student Researcher* March 2024 – Present

- Performed runtime and memory analysis; enhanced data loading efficiency of conditional GAN by 100 times through **Dask** Delayed objects while ensuring compatibility with **PyTorch** tensor datasets; enhanced pipeline robustness via rigorous **PyTest** coverage, **Dockerized** deployments, and transformer validation
- Refined and tested novel jailbreak techniques with ollama and HarmBench, AgentDojo datasets; co-authored EMNLP 2025 submission on jailbreaking multi-agent systems built on **AutoGen**

Upsilon Pi Epsilon (UPE) | *Web Development Officer* January 2025 – Present

- Maintained and enhanced an existing website serving **2000+ CS students**, contributing full-stack improvements using **Django** for backend logic, **JavaScript** for interactivity, and **HTML/CSS** for UI refinements
- Refactored and supported the test bank system to ensure secure, reliable access for UPE members; applied **Linux DevOps** practices for server management, deployment, and performance monitoring

UCLA Student Media | *Full-Stack Developer* January 2025 – Present

- Maintained website used by **2500+ applicants** across Student Media organizations (e.g., *Bruinwalk*), implementing front-end interfaces in **JavaScript/HTML/CSS** and backend/auth in **Django** with **PostgreSQL** database
- Enhanced applicant data workflows by implementing backend endpoints with dynamic filtering capabilities, reducing query times by **30%** and streamlining the review process for hiring managers

Cyber Summer Research Fellowship | *Cybersecurity Researcher* July 2024 – September 2024

- Developed convolutional neural network using **PyTorch** to detect image steganography (LSB, PVD, DCT, FFT), applying transfer learning through established models (EfficientNet), achieving over **87% accuracy** in multiclass classification on over **33,000 image dataset**
- Co-authored Deep Learning for Steganographic Image Detection, accepted by 2024 IEEE MIT URTC Conference

Glocol Networks | *Software Intern* June 2022 – July 2022

- Engineered real-time occupancy detection for system mass transit vehicles using **Python** algorithms, Bluetooth signals with Raspberry Pi devices to collect beacon data from mobile devices, deployed on **AWS IoT Greengrass Lambda**
- Extracted Amtrak train departure and arrival times and displayed on frontend with **JavaScript, Vue.js, HTML/CSS**

Projects

ACM AI | *Python, PyTorch, Pandas, Pillow, NumPy* January 2024 - Present

- Led a 3-member team to design Wasserstein Generative Adversarial Network (**WGAN**) using **PyTorch**, generating and classifying emojis from dataset of over 10,000 images; performed hyperparameter tuning and model evaluation
- Implemented **GPT-2** from scratch with PyTorch, developing embeddings, multi-head attention, and stochastic decoding

BruinBowl | *MongoDB, Express, React, Node* March 2024 - June 2024

- Collaborated with 6 students to design interactive web application on UCLA trivia, using Git for version control
- Designed user and gameplay functionality using **MongoDB, Express, React, Node**; created secure user login and authentication alongside features scalable for hundreds of users: question search, real-time leaderboards, and user profiles

Technical Skills

Languages: Python, C/C++, JavaScript, HTML/CSS, Java, SQL, Bash

Technologies/Frameworks: MongoDB, Express.js, React, Node.js, PyTorch, Git, Docker

Certifications : Machine Learning Specialization (Deeplearning.ai/Stanford)